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Knowledge Enhancement and Temporal Aware for Multi-Behavior Contrastive Recommendation

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A well-designed recommender system can accurately learn the embeddings of users and items, reflecting the unique preferences of users. Traditional recommendation techniques usually focus on modeling the singular type of behaviors between users and items. However, in many practical recommendation scenarios (e.g., social media, e-commerce), there exist multi-typed interactive behaviors in user-item relationships, such as click, tag-as-favorite, and purchase in online shopping platforms. Thus, how to make full use of multi-behavior information for recommendation is of great importance to the existing system, which presents challenges in two aspects that need to be explored: (1) Utilizing users' personalized preferences to capture multi-behavioral dependencies; (2) Dealing with the insufficient recommendation caused by sparse supervision signal for target behavior. In this work, we propose the Knowledge Enhancement Multi-Behavior Contrastive Learning (KMCL) framework, including two Contrastive Learning tasks and three functional modules to tackle the above challenges, respectively. In particular, we design the multi-behavior learning module to extract users' personalized behavior information for user-embedding enhancement and utilize knowledge graph in the knowledge enhancement module to derive more robust knowledge-aware representations for items. In addition, in the optimization stage, we also model the coarse-grained commonalities and the fine-grained differences between multi-behavior of users to further improve the recommendation effect and propose a joint training paradigm to enhance the learning effect of KMCL in the joint learning module. Besides, we also considered how to make full use of temporal signals to enhance the effectiveness of multi-behavior recommendations in scenarios with time information and designed a novel encoder to address this issue. Extensive experiments and ablation tests on the three real-world datasets indicate that our KMCL outperforms various state-of-the-art recommendation methods and verify the effectiveness of our method.

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1 Introduction

Recommender systems aim to provide items for users personalized based on their preferences, which have emerged as critical components to alleviate information overloading in various online applications [2, 9, 34, 37]. How to accurately capture users' preferences from their behaviors is the key to personalized recommendations. **Collaborative Filtering (CF)** techniques are traditional and effective, which forecast user preference from observed behavior data. The core idea of CF paradigm is to factorize user-item interactions into latent representations and excavate the similarity preference of individual users based on vectorized user/item representations to make recommendations [11].

With the recent boosting of deep learning, neural networks have been utilized to augment CF architecture. For instance, in 2017, He et al. and Xue et al. introduced the Multi-layer Perceptron and proposed NCF [11] and DMF [52], enabling that the CF could handle the non-linear interaction. In 2018, by stacking multiple autoencoder networks [28], Du et al. designed a method to realize the high-dimensional sparse user behavior data mapping into the low-dimensional dense representations [4]. Inspired by the recent success of GNN, other research works model user-item interactions as graphs to exploit the high-order user-item relations. These models perform message passing and neighborhood-based feature aggregations over the interaction graph to generate node-level embeddings (e.g., NGCF [36] and GCN [58]).

Nevertheless, existing recommendation methods predominantly focus on singular user-item interactions, real-world scenarios demand the simultaneous handling of multidimensional behavioral patterns (e.g., clicks, favorites, purchases) and multimodal item representations (e.g., visual product aesthetics, textual reviews, temporal usage contexts). Traditional multi-behavior approaches, such as MATN [39], excel at modeling cross-behavior dependencies (e.g., click-to-purchase transitions) but often neglect how multimodal content (e.g., product images or review semantics) contextualizes behavioral intent. Conversely, multimodal frameworks like DiffMM [12], PrmptMM [38] and FETTLE [17] align visual-textual features but fail to capture behavioral granularity—for instance, overlooking how repeated “favorite” actions on stylistically similar items (visual modality) correlate with eventual purchases (target behavior).

This dichotomy underscores their complementary potential. On e-commerce platforms, users exhibit intent through both behavioral diversity (e.g., browsing jackets tagged as “trendy”) and multimodal engagement (e.g., favoring items with specific color schemes in images or textual descriptors like “lightweight”). While ATRank [24] sequences behavioral interactions, it ignores how visual-textual alignment (via multimodal **Contrastive Learning (CL)**) could resolve ambiguities—e.g., distinguishing casual clicks on winter coats from intentional purchases driven by “waterproof” keyword matches in reviews. A synergistic integration could bridge this gap, where behavioral sequences inform temporal intent, and multimodal signals disambiguate preference semantics.

So how to accurately capture the dependencies between behavioral diversity and distinguish the differences between users is a challenging but valuable work. For instance, ATRANK [59] focuses on the feature interactions between different user behaviors in sequence recommendation and attaches importance to a single sequence view of user's historical behavior. By utilizing self-attention mechanism, MATN [42] realizes the encoding of pairwise correlations between different types of behavior to recommend target behaviors.

Although the above methods are indeed effective, there are still two limitations in practical research: (1) target behavior data sparsity: Most recommender systems utilize supervised information for training, so it is pivotal to have enough target behavior data for the accuracy of prediction results. Unfortunately, in real-world scenarios, interactions observed under target behavior are often sparse compared to other types of interactions. It is obvious that the data of people purchasing on e-commerce platforms are much lower than the data of clicks, which leads to the direct integration of certain types of behavior that will sacrifice recommendation performance. (2) Individualized behaviors diversity: The dependencies between different types of interactions are complex, while the interdependencies between multiple types of behavior vary from user to user, thus leading to two problems. One is how to model the coarse-grained commonalities between different behaviors and another is how to model the fine-grained differences between multi-behavior of users. In practical scenarios, individuals exhibit different behavior patterns according to their own habits. For example, some users will frequently perform tag-as-favorite but only make sporadic purchases, while others may be not used to the operation of add-to-cart. Therefore, building an efficient information transfer framework to extract sufficient multi-behavioral patterns is crucial.

Recently, many studies have introduced CL into recommendation and achieved certain results [41, 55]. We find that CL is well suited for modeling multi-behavior information to explore the coarse-grained commonalities and fine-grained differences between user-item relationships. Meanwhile, **Knowledge Graphs (KG)** have been widely used in recommendation and become a popular method for information completion. Hence, inspired by work related to CL and KG, we propose a **Knowledge Enhancement Multi-Behavior Contrastive Learning Recommendation (KMCLR)** framework to tackle above limitations [51]. Specifically, KMCLR consists of three modules: multi-behavior learning module, knowledge enhancement module, and joint learning module, which are utilized to learn user-item representations from multiple perspectives. Our multi-behavior learning module captures cross-type behavior dependencies and utilizes additional supervision signals from different behaviors to complement and enhance user-level information. While our knowledge enhancement module proposes a relation-aware knowledge aggregation mechanism, which distills additional KG information to guide the data expansion of cross-view signals and achieves item-level information enhancement. Finally, two different CL tasks and a loss paradigm are proposed to maximize the differences between representations of users while minimizing the gaps between different behaviors of the same user.

In addition to the aforementioned work, to consider the impact of common sequence relationships and temporal signals on recommendation results in real-life scenarios, we also conduct expansion work. In the extended work, we design an additional multi-behavior encoder based on temporal signals for the KMCLR and propose a collaborative attention mechanism based on behavior categories and time periods. Through comparative experiments with other methods, it can be concluded that this encoder can dynamically model user preferences based on different contextual information and solve the problem of inaccurate recommendations caused by unified modeling of information in the entire time period.

In summary, the contributions of our work are highlighted as follows:

- We propose a new framework KMCLR for recommendation, which enhances the information from the user level and item level by combining multi-behavior messages with KG messages, emphasizing the modeling of user-item information from different perspectives, solving the issue of target behavior data sparsity.
- In our KMCLR framework, we propose two CL tasks and a loss paradigm, which can better model coarse-grained commonalities between behaviors and fine-grained differences between users to improve the recommendation performances. To the best of our knowledge, this is the first time that multi-behavior CL and KGCL are jointly introduced into recommendation.
- To make the model more suitable for industrial scenarios, we also design a novel multi-behavior encoder based on the temporal information in real business. This encoder can be used as part of KMCLR or applied alone in recommendations, effectively achieving dynamic modeling of the evolution of individual preferences.
- We conduct diverse experiments on three real-world datasets to show that our framework outperforms compared to various state-of-the-art recommendation methods. Furthermore, the ablation study is performed to better understand the model design of KMCLR and justifies the effectiveness of our key components.

2 Preliminaries

In our studied task, we let $\mathcal{U}$ and $\mathcal{V}$ represent the set of users and items: $\mathcal{U} = \{u_1, \dots, u_i, \dots, u_I\}$ and $\mathcal{V} = \{e_1, \dots, e_j, \dots, e_J\}$, where I and J represent the number of users and items, respectively. We give detailed definitions of the key notions in our KMCLR as follows:

User-Item Multi-Behavior Interaction Graph. Since there are various types of behavior in user-item interactions, a multi-behavior interaction graph is defined as: $G_u = (\mathcal{U}, \Theta, \mathcal{V})$, where $\Theta = \{\theta^1, \dots, \theta^k, \dots, \theta^K\}$ is the set of edge representing K types of behavior. Specifically, if user u_i and item v_j have an interaction under the type of behavior k , then $\theta_{i,j}^k = 1$, and $\theta_{i,j}^k = 0$ otherwise.

Item-Item Relation KG. Considering the richness of external knowledge of items, we utilize the traditional triple method to define KG: $G_v = (H, R, T)$, where H denotes head-entity set, R is relation set, and T represents tail-entity set. Among them, the combination of elements with the same index of these three sets forms a triple (h, r, t) , which reflects the semantic relationship r between the head and tail entity h and t . In particular, we refer to the form of work [43] to build G_v . And the head-entity and tail-entity set are the set of candidate items.

Task Formulation. Based on the above definitions, we formulate the studied task as:

Input. The user-item multi-behavior interaction graph $G_u = (\mathcal{U}, \Theta, \mathcal{V})$ and the item-item relation KG $G_v = (H, R, T)$.

Output. A prediction function which infers the interaction probability $y_{i,j}^k$ of user u_i and item v_j under the k -type behavior.

3 Methodology

The overall architecture of the KMCLR model is shown in Figure 1. Our model mainly consists of three parts: multi-behavior learning module, knowledge enhancement module, and joint learning module. Two types of CL tasks are proposed to capture the multi-behavior interaction feature and item-layer knowledge feature.

3.1 Multi-Behavior Learning Module

In this module, the user's multi-behavior interaction graph is fed into multi-behavior encoder. We construct user single-behavior representations through this encoder. KMCLR optimizes the

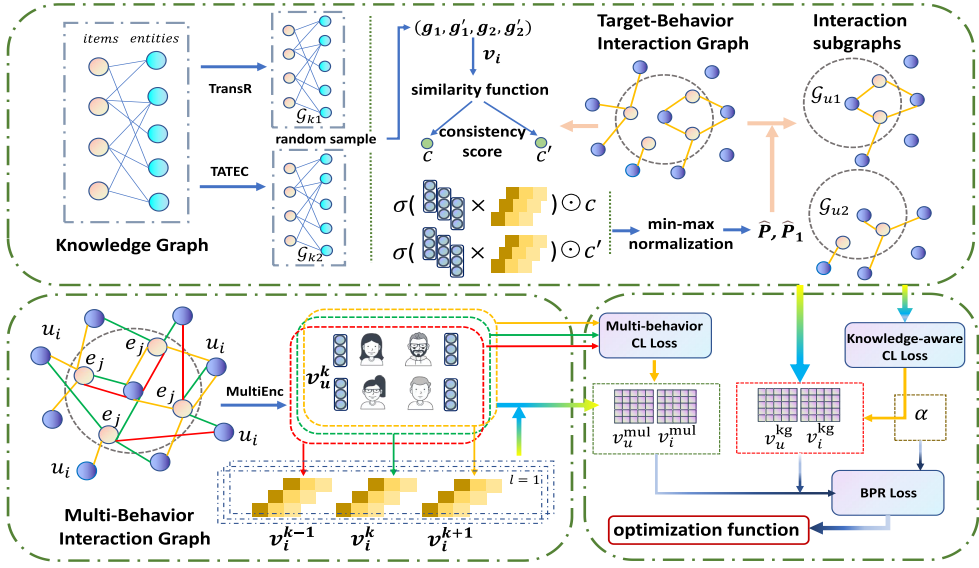


Fig. 1. The model architecture of KMCLR framework.

user-item embeddings according to the contrastive loss of other behaviors with the target behavior. This process leverages multi-behavioral information to enhance user-level embedding representations.

3.1.1 Multi-Behavior Information Encoding and Aggregation. Graph-based recommendation is usually performed on the user-item graph constructed by all behaviors, which can consider interactions among global multi-hops. To take full advantage of multi-behavior interaction graphs while injecting higher-order connectivity into multi-way relations, we develop a behavioral context-aware information encoder for learning single-behavior user representations:

$$v_u^k, v_i^k = \text{MultiEnc}^k(G_u, u, k), \quad (1)$$

where v_u^k and v_i^k indicate the obtained representations of user u and item i under the behavior k . $\text{MultiEnc}^k(\sim)$ is an optional multi-behavior encoder. Inspired by the lightweight architecture of LightGCN [10] and advanced information propagation framework MB-GMN [45], we utilize the following formula as $\text{MultiEnc}^k(\sim)$ to calculate the node (consist of user and item) representations:

$$v_u^{k,(l+1)} = \sum_{i \in N_u^k} v_i^{k,l}; \quad v_i^{k,(l+1)} = \sum_{u \in N_i^k} v_u^{k,l}, \quad (2)$$

where N_u^k, N_i^k are the neighboring nodes of user u and item i . $(l+1)$ means that the encoding operation is performed on the $(l+1)$ th layer of the neural network. Especially, we utilize the raw user and item embedding to define $v_u^{k,0}$ and $v_i^{k,0}$. Since single-behavior representations may suffer from data sparsity issue, we propose to perform cross-type multi-behavior embedding aggregation, applying multiple layers of transformation matrices while preserving contextual information. The specific formulas are as follows:

$$v_u = \text{PreLU}((v_u^0 || \dots || v_u^l || \dots || v_u^L) \times W^l), \quad (3)$$

$$v_u^l = \sigma(W^u \times \text{mean}(v_u^{1,l} \oplus \dots \oplus v_u^{K,l} \oplus \dots \oplus v_u^{K,l})). \quad (4)$$

$W^u \in \mathcal{R}^{d \times d}$ and $W^l \in \mathcal{R}^{L \cdot d \times d}$ are trainable transformation parameters, with d as latent dimension and L as the number of neural network layers. σ is defined as sigmoid activation function. Similar operations are applied for item aggregation.

3.1.2 Multi-Behavior CL. Influenced by the CL strategy, we hope to utilize multiple views to construct different representations and regard them as auxiliary signals to alleviate the target-behavior data sparsity. Meanwhile, we try to minimize the difference between multi behaviors of the same user and maximize the gap between each user through the contrastive strategy. Therefore, a naive idea is to treat different behaviors as different views. To be precise, we perform CL on user embeddings between behaviors and realize cross-type behavioral dependency encoding through different embedding combinations, completing user-level information enhancement.

One of the keys to CL is to extract valid positive-negative pair samples from different views. In our KMCLR, we randomly select different behaviors of user x as positive pairs, while views between users x and y are sampled as negative pairs. Therefore, according to the user embedding generated before, we further get the positive and negative sample pairs: $\{v_x^k, v_x^{k'}\}$ and $\{v_x^k, v_y^{k'}\}$, where $x, y \in \mathcal{U}$ and $x \neq y$. According to work [25], we define a self-supervised contrastive loss based on InfoNCE loss as follows:

$$\mathcal{L}_{MulCL} = \sum_{k'=1}^K \sum_{x \in \mathcal{U}} -\log \frac{\exp(s(v_x^k, v_x^{k'})/\tau)}{\sum_{y \in \mathcal{U}} \exp(s(v_x^k, v_y^{k'})/\tau)}. \quad (5)$$

τ is the parameter to control the smoothness of the softmax curve. $s(\sim)$ denotes the pairwise distance function to reflect the similarity of pos-neg pairs. Through the multi-behavior CL task, KMCLR can learn more robust user representations, capture the underlying dependencies between target behavior and other behaviors, and distinguish personal preferences among different users.

3.2 Knowledge Enhancement Module

The KG is input into the knowledge enhancement module as auxiliary information. KMCLR utilizes different graph embedding techniques, focusing on the rationality and semantic information in the KG to obtain different node representations. Based on each graph embedding representation, an item score evaluation method from the user's perspective is proposed to construct user-item embeddings by preserving low-noise item information. These processed data will be further utilized to optimize item-level embedding representations according to CL task.

3.2.1 Item-Side Information Encoding and Aggregation. As the supplement to item-side information, KG contains rich relational representations on its edges. Under most circumstances, the importance of neighbor nodes in the KG is different. To reflect this difference, inspired by the graph attention mechanism [32], we design a method of aggregating neighbor nodes by attention weight and obtain the representations of items:

$$\mathcal{W}_{i,r,e} = \frac{f_{att}(v_i, v_r, v_e)}{\sum_{e' \in \mathcal{N}_i} f_{att}(v_i, v_r, v_{e'})}, \quad (6)$$

$$\begin{aligned} f_{att}(v_i, v_r, v_e) &= \exp[\sigma(W_1(v_i || v_r || v_e)) + b_1], \\ v_i^{l+1} &= \sigma(W_2(v_i^l + \sum_{e \in \mathcal{N}_i} \mathcal{W}_{i,r,e} v_e) + b_2). \end{aligned} \quad (7)$$

where $\mathcal{W}_{i,r,e}$ denotes the weight of item i to the neighbor node $e \in \mathcal{N}_i$ under the relation r . Both $W_1, W_2 \in \mathcal{R}^{3d}$ and $b_1, b_2 \in \mathcal{R}$ are trainable parameters. v_i^{l+1} is the embedding of item i after $(l+1)$ th aggregations. Here, we use v_i^0 to denote the raw item embedding.

3.2.2 Knowledge-Based Enhancement and Augmentation. As mentioned before: we hope to construct multiple appropriate views for CL. Therefore, two different approaches are proposed to generate knowledge representations, which are the translational distance model TransR [19] and the semantic matching model TATEC [8]. Respectively, the two representations focus on the rationality and potential semantics of the triples in KG, so as to obtain the knowledge augmented graphs with different focus points. The specific formulas are as follows:

$$f_{td}(h, r, t) = -||M_r v_h + v_r - M_r v_t||_2^2, \quad (8)$$

$$f_{sm}(h, r, t) = v_h^T M_r v_t + v_h^T v_r + v_t^T v_r + v_h^T D v_t. \quad (9)$$

$M_r \in \mathcal{R}^{d \times d}$ is a projection matrix from the entity space to the relation space of r , while D is a diagonal matrix shared across all different relations. v_h, v_r, v_t stand for triple embeddings in KG. So we initialize to get two different representations of the graph: $\mathcal{G}_{k1}$ and $\mathcal{G}_{k2}$. Then, two different loss functions are exploited to optimize the node representations of KG:

$$\mathcal{L}_{td} = \sum_{(h,r,t,t') \in \mathcal{G}_{k1}} -\ln \sigma(f_{td}(h, r, t') - f_{td}(h, r, t)), \quad (10)$$

$$\mathcal{L}_{sm} = \sum_{(h,r,t,t') \in \mathcal{G}_{k2}} -\ln \sigma(f_{sm}(h, r, t') - f_{sm}(h, r, t)), \quad (11)$$

where (h, r, t') is the triple that does not exist in KG. Through the above methods, KMCLR further realizes the enhancement of knowledge representation from different perspectives

Although the introduction of KG can complement the recommendation information, it also introduces some noise. Therefore, inspired by work [14, 53, 63], if the node representation obtained by different knowledge subgraphs is similar, it denotes that the node is not sensitive to structural changes with strong noise immunity. We propose a simple method to compute graph structure consistency, reducing the effect of noisy information:

$$c = s(g_1(v_i), g'_1(v_i)); \quad (g_1, g'_1 \in \mathcal{G}_{k1}^{sub}), \quad (12)$$

$$c' = s(g_2(v_i), g'_2(v_i)); \quad (g_2, g'_2 \in \mathcal{G}_{k2}^{sub}). \quad (13)$$

Here, (g_1, g'_1, g_2, g'_2) are subgraphs sampled from $\mathcal{G}_{k1}$ and $\mathcal{G}_{k2}$ by setting different seeds. $s(\sim)$ is the similarity function to calculate the consistency parameter $c \in \mathcal{R}^{1 \times M}$ for all item nodes. Specifically, we obtain the node representations v_i of subgraphs by Equation (7). Generally speaking, the stronger stability and noise resistance of a node, the higher its corresponding score c , which is used as guiding information to optimize the user-item interaction graph.

Stable nodes benefit to resist noise, but not enough to reflect user preferences. Some users may prefer to interact with nodes of low stability. So, after obtaining the consistency score of nodes in KG, we attempt to adjust the score from user's perspective and utilize the score to guide the selection of edges in interaction graph. We designed the following sampling probability calculation formulas:

$$P_{u,i} = \sigma(v_u^T v_i) \odot c, \quad (14)$$

$$\hat{P}_{u,i} = (1 - \text{Min_Max}(P_{u,i}))a + \text{Min_Max}(P_{u,i})b. \quad (15)$$

$P_{u,i}$ is the initial sampling probability based on user's interactions and consistency score c , while $\odot$ is the element-wise multiplication with a broadcast mechanism. KMCLR performs min-max normalization function $\text{Min_Max}(\sim)$ to transform $P_{u,i}$ into interval $[a,b]$ to get final sampling probability of interaction graph. According to this probability, the edge retention in the interaction graph is determined, obtaining two user-item subgraphs $\mathcal{G}_{u1}$ and $\mathcal{G}_{u2}$.

3.2.3 Knowledge-Aware CL. After the above steps, KMCLR obtains two pairs of perspective graphs with different emphasis information: $(\mathcal{G}_{u1}, \mathcal{G}_{k1})$ and $(\mathcal{G}_{u2}, \mathcal{G}_{k2})$. According to Equation (7), the representations of items are encoded to preserve the information of KG. Then, similar to the multi-behavior learning module, we utilize LightGCN to perform message propagation and calculate the embeddings of user-item graph. The specific formula is the same as Equation (2).

Based on the two pairs of perspectives mentioned above, KMCLR performs CL on view-specific node representations, which treats the same node embeddings from different views as positive pairs and different nodes from different views as negative pairs. Similar to Equation (5), we define the following formula:

$$\mathcal{L}_{KCL} = \sum_{x \in \mathcal{U} \cup \mathcal{V}} -\log \frac{\exp(s(v_x^1, v_x^2)/\tau)}{\sum_{y \in \mathcal{U} \cup \mathcal{V}} \exp(s(v_x^1, v_y^2)/\tau)}. \quad (16)$$

Through the knowledge-aware CL task, KMCLR can supplement the item-side information from the external KG, reduce the impact of external noise, and alleviate the issue of data sparsity to a certain extent.

3.3 Joint Learning Module

Through the above modules, we obtained two types of user-item embeddings generated in different ways, one using multi-behavior information and the other using KG information. These embeddings obtained from different perspectives contain messages with different emphases, which require to be combined in a reasonable way. Therefore, we assign a combination weight α and explore the optimal combination result through experiments. We can get:

$$v_u = (1 - \alpha)v_u^{mul} + \alpha v_u^{kg}, \quad v_i = (1 - \alpha)v_i^{mul} + \alpha v_i^{kg}. \quad (17)$$

During the optimization phase, following the classical ranking model [27], KMCLR leverages the **Bayesian Personalized Ranking (BPR)** recommendation loss as the main task loss to further optimal parameters:

$$\mathcal{L}_{BPR} = - \sum_{u \in \mathcal{U}} \sum_{i \in \mathcal{N}_u} \sum_{j \notin \mathcal{N}_u} \text{In}\sigma(v_u^T v_i - v_u^T v_j), \quad (18)$$

where $\mathcal{N}_u$ denotes the set of observed interactions of users u .

After defining all three losses, a training strategy paradigm is proposed to better learn information from different views and further optimize the model. Specifically: (1) First, we will train the two CL models separately in the multi-behavior learning module and the knowledge enhancement module, learning different initial parameter spaces of the two modules on separate training data. (2) Then, the two-part trained node embeddings are concatenated with the combined weight α . We calculate the main model BPR loss and CL loss for the new node embeddings as the final optimization step of KMCLR. It should be noted that after study, we decided to utilize LightGCN in the multi-behavior learning module as our main model. The specific training paradigm is as follows:

$$\begin{aligned} \text{argmin} \triangleq & \text{main_opt}\{\text{mul_opt}(L^{mul}, V^{mul}), \\ & \text{kg_opt}(L^{kg}, V^{kg}), V^{main}\}. \end{aligned} \quad (19)$$

Here _opt is the parameter optimizer, L and V are the loss and node representation set of the corresponding module, respectively. Among them, for the loss L , we give the following definition:

$$L = \mathcal{L}_{BPR} + \lambda_1 \mathcal{L}_{cl} + \lambda_2 \|\Omega\|_2^2, \quad (20)$$

where Ω is the learnable model parameters, while λ_1 and λ_2 are utilized to control the weights of self-supervised signals and regularization terms.

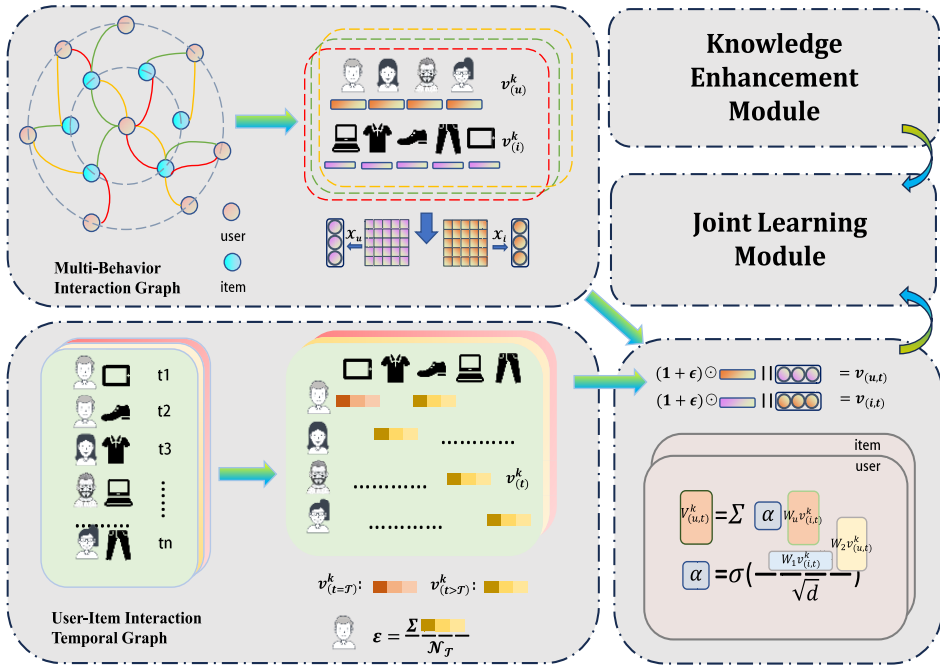


Fig. 2. The architecture of KMCLR with temporal-aware multi-behavior encoder.

3.4 How to Utilize Temporal Signals

Actually, temporal signals are very crucial in recommendation systems, as they reflect the evolution of user preferences from a dynamic perspective [56]. If we discard the temporal information and merely utilize the interactive data, which will make it difficult for systems to distinguish between individual historical preferences and current preferences, thereby interfering with the accuracy of recommendations. Recently, many works have been devoted to integrating temporal information with recommendation systems, e.g., NARM [16], GC-SAN [48], and SR-GNN [40]. The above methods belong to the field of sequence recommendation, which usually constructs interaction data as session sequences, and the position of item in the session contains temporal information to a certain extent.

However, for multi-behavior recommendation, most existing methods only consider how to accurately model the cross-behavior dependencies, while ignoring the temporal signals that are sufficient to reflect the user's continuous preferences. In fact, personal preferences and behavioral paradigms dynamically evolve with different contexts [44], which result in certain historical preferences and behavioral habits not accurately profiling current user portraits. Therefore, if the contribution of all interactive data to the final recommendation is uniformly averaged without considering the temporal information, it will damage the feature expression ability of the recommendation system.

So in view of our previous research work [50], this article design a *temporal-aware multi-behavior encoder* to address the above issue. It should be noted that this encoder is a supplement and extension to the KMCLR model, which can replace the *multi-behavior encoder* in Section 3.1.1 under appropriate circumstances to achieve better results. It can also be utilized as a separate multi-behavior model for recommendation scenarios that require processing of temporal signals. Figure 2 shows the structure of KMCLR based on the temporal-aware multi-behavior encoder, whose output can be used as input for multi-behavior contrastive tasks for joint training.

3.4.1 Supplementary Input. In this task, the user-item multi-behavior interaction graph G_u is also the input of the model. We give detailed definitions of another key input data as follows:

User-Item Interaction Temporal Graph. For each piece of interaction data, we extract its key attribute information to construct the graph $G_t = (\mathcal{I}, \Theta, \mathcal{T})$, where $\mathcal{I}$ is the id set of interaction parties, Θ is the set of behavior type mentioned in G_u , and $\mathcal{T}$ denotes the temporal signal of each interaction.

3.4.2 Temporal Multi-Behavior Information Encoding. To fully extract node features in graph for mining the latent expression containing user potential preferences and item attribute information, we design a general context encoder that jointly considers temporal signals and behavioral information to construct node embeddings. Inspired by the advanced positional embedding method in Transformer [31], the following formulas are adopted to obtain each type of node representation:

$$v_{2i}^k = \sin\left(\frac{\phi(t^k)}{10000^{\frac{2i}{d}}}\right); \quad v_{2i+1}^k = \cos\left(\frac{\phi(t^k)}{10000^{\frac{2i}{d}}}\right); \quad v_{(t)}^k = v^k \cdot W_t; \quad (21)$$

$$v_{(u)}^{k,(l+1)} = \sum_{i \in \mathcal{N}_u^k} \frac{v_{(i)}^{k,l}}{\sqrt{\mathcal{N}_u^k} \sqrt{\mathcal{N}_i^k}}; \quad v_{(i)}^{k,(l+1)} = \sum_{u \in \mathcal{N}_i^k} \frac{v_{(u)}^{k,l}}{\sqrt{\mathcal{N}_i^k} \sqrt{\mathcal{N}_u^k}}, \quad (22)$$

where $v_{(u)}^{k,l}$ and $v_{(i)}^{k,l}$ denote the embeddings of user u and item i under behavior k on the l th layer, while $v_{(t)}^k$ indicates the temporal representation of all interaction related to behavior k . And $\mathcal{N}_u^k, \mathcal{N}_i^k$ are the sets of neighbors of the corresponding node under k th behavior. $W_t \in \mathcal{R}^{2d \times d}$ is a trainable parameter for temporal signals transformation. Motivated by the work of KHGT [43], the function $\phi(\sim)$ is defined as a time-period function to map each interaction timestamp t^k to the time slot, where t^k is indexed by $2i$ and $2i+1$ (representing odd and even positions, respectively).

3.4.3 Temporal-Aware Attention. Comparing with Section 3.1.1, it should be noted that in addition to the conventional embeddings $v_{(u)}^k$ and $v_{(i)}^k$, we also constructed additional interactive representations $v_{(t)}^k$ with temporal information, which can be used in time-aware scenarios. And then to achieve accurate modeling of input concatenated features, we designed a novel attention vector based on temporal information, which can effectively distinguish individual projections that depend on historical interactions, and construct more robust preference expressions. The specific formulas are as follows:

$$v_{(u,t)}^k = (1 + \varepsilon) \odot v_{(u)}^k; \quad \varepsilon = \frac{\sum_{t \in \mathcal{N}_t} v_{(t)}^k}{\mathcal{N}_t}. \quad (23)$$

$\mathcal{N}_t$ is the interactions set of user u . Since the value of ε will vary according to the personalized preference evolution trend, it reflects the user's attention to time series signals. According to the research of Kumar et al. [15], we exploit $(1 + \varepsilon)$ to scale the original user embedding $v_{(u)}^k$. When $\varepsilon = 0$, $v_{(u,t)}^k$ degenerates into node representation without temporal information. In this case, we can change the value of ε by setting time intercepts to extract data in different periods, thereby distinguishing the local preferences of individuals in each time segments. Through the above operations, the model can effectively identify the interactive attitudes of different behavior types toward implicit time signals. For example, for the "click" behavior, we consider that due to its large data volume and high aimlessness, long-term data contribute less to current preferences than short-term data, so setting an intercept close to the current time point will achieve better results. While for the "tag-as-favorite" behavior, we can let the model consider the existence of more previous interaction information due to its higher stability.

Since the individual's behavioral paradigm has step changes (similar to edge features in the CV field), this is difficult to discover in recommendation systems which do not contain sequence relationships or temporal information, while these step signals contain key preference transformation information. There is previous work dedicated to studying edge signals in graphs. Simonovsky and Komodakis [29] successfully merged edge features through the spectral-based GCN. Battaglia et al. [1] designed a novel basic framework for processing edge signals based on the GNN model. Encouragingly, in 2020, the TGAT model [49] proposed to utilize message propagation to process edge signals in temporal graphs through a natural way. In view of the previous analysis and the impact of TGAT, we rewrite Equation (23) as follows:

$$v_{(u,t)}^k = (1 + \varepsilon) \odot v_{(u)}^k || \mathcal{X}_u; \quad \varepsilon = \frac{\sum_{\mathcal{T}} v_{(t>\mathcal{T})}^k}{\mathcal{N}_{\mathcal{T}}}, \quad (24)$$

where $\mathcal{T}$ is the time intercept mentioned before, and $\mathcal{X}_u$ is the user's feature vector of interaction behavior matrix. Through the above formula, the model can fuse edge signals with node representations through information aggregation and propagate them to the next layer of the network. Similar operations are also applied to item side.

3.4.4 Temporal Multi-Behavior Information Aggregation. In order to mine the dependence of neighbor temporal information on the target node while maintaining behavioral heterogeneity, we propose a message aggregation paradigm to combine temporal cooperation signals with multi-behavior information. For a single behavior k and each layer of neural network, the specific mechanism is as follows:

$$V_{(u,t)}^k = \sum_{i \in \mathcal{N}_u} \alpha_{(i,t)}^u W_u v_{(i,t)}^k; \\ \alpha_{(i,t)}^u = \sigma\left(\frac{(W_1 v_{(i,t)}^k)^T (W_2 v_{(u,t)}^k)}{\sqrt{d}}\right), \quad (25)$$

where $\alpha_{(i,t)}^u$ represents the importance weight of item i to interactive object user u at time t . $\sigma(\sim)$ is the softmax activation function. Both $W_u \in \mathcal{R}^{2d}$ and $W_1, W_2 \in \mathcal{R}^d$ are trainable parameters. $\alpha_{(i,t)}^u$ is actually the collaborative attention of the historical interactive information (u, i, t) to the target embedding $v_{(u,t)}^k$ based on temporal inference. Compared with the self-attention, this collaborative attention unifies neighbor information and temporal causal inference facts, which is a more effective mechanism for capturing potential preferences [5].

Although the KMCLR framework has demonstrated robust performance in multi-behavior recommendation, it does have certain limitations, particularly in capturing the dynamic evolution of user preferences across different time periods. To address this, our extended method introduces a temporal-aware multi-behavior encoder, which significantly enhances the model's ability to dynamically model user preferences and behavioral paradigms. This new encoder, as detailed in Section 3.4, allows us to incorporate temporal signals effectively, thus overcoming the previous limitations and providing a more accurate and nuanced recommendation.

After constructing the individual representation of a single behavior, the model obtains unified user representation through the aggregation mechanism between cross-type behaviors. For the specific formula, please refer to Equations (3) and (4). Through the above steps, the model realizes dynamic modeling of individual context features based on temporal signals and acquires embedding $V_{(u,t)}^k$ containing user's preference evolution information. The same operations are applied to obtain the item-side embedding.

3.5 Model Complexity Analysis

We analyze the complexity of KMCLR from the following aspects: (1) In the multi-behavior learning module, the cost of lightweight neural architecture is $O(L \times K \times |\Theta^+| \times d)$ with L network layers, K types of behavior, $|\Theta^+|$ edges set of interactions, and d hidden dimensionality. The cross-type of behavior aggregation costs $O(L \times K \times (I + J) \times d)$. The complexity to calculate InfoNCE loss is $O(B \times K \times |\Theta^+| \times d)$, where B is the sample size in an epoch of training. (2) In the knowledge enhancement module, the operations of information aggregation take $O(|R| \times d)$ with $|R|$ as the number of relations in G_v . For the generation of each pair of perspective graphs, KMCLR costs $O(|\Theta^+| \times d^2 + |\Theta^+| \times d \times L)$. The complexity of InfoNCE-based loss calculation is $O(B \times d \times (I + J))$. Compared to existing models, KMCLR improves performance by slightly sacrificing complexity. (3) In the temporal-aware encoder, the architecture spends $O(L \times K \times (I + J) \times d^2)$ to implement the calculation of user/item embeddings containing temporal signals. Compared to conventional transformer-based models, we have achieved more accurate node representation with the same complexity through structural improvements.

4 Experiments

In this section, we evaluate the performance of KMCLR through extensive experiments and answer the following questions: (RQ1) How does KMCLR perform compared to other recommendation methods? (RQ2) What are the effects of different modules in KMCLR on performance? (RQ3) How does KMCLR effectively alleviate data sparsity issue for recommendation? (RQ4) How different hyperparameter settings affect the performance? (RQ5) How does KMCLR alleviate the noise issue caused by extra information? (RQ6) How does our proposed temporal-aware multi-behavior encoder perform?

4.1 Datasets for Experiments

We validate the effectiveness of KMCLR with experiments on three real-world datasets. *Yelp*: This dataset is collected by well-known business review web sites in the United States, covering restaurants, shopping malls, hotels and other fields, usually used for business venue recommendation. According to the settings of [43], we considered four behaviors (i.e., *dislike*, *neutral*, *tip*, *like*). *Tmall*: It is collected by one of the largest E-commerce platforms in China. In this dataset, user interactions include *page view*, *tag-as-favorite*, *add-to-cart*, and *purchase*. *Retail*: It is a dataset about online retail scenarios with explicit multi-type user-item interactions. We extracted the following three behaviors as training data, i.e., *tag-as-favorite*, *add-to-cart*, and *purchase*. Table 1 presents the detailed statistics of the above datasets. Furthermore, referring to previous studies, we cite the KG constructed in work [43] during the experiment.

4.2 Methods for Comparison

We compare KMCLR with various baselines from different research lines as follows:

CF-Based Method.

- *BPR* [27]. It is a traditional CF method for personalized ranking of items with pairwise ranking loss.
- *LightGCN* [10]. It uses GNN for CF and simplifies convolution operations between message propagation.
- *AutoRec* [28]. It is an autoencoder-based model that projects user-item interactions into a hidden low-dimensional space.
- *NGCF* [36]. It is a message passing framework based on graph neural network CF, which utilizes high-order connectivity for aggregation.

Table 1. Statistics of Datasets

Dataset	Users	Items	Interactions	Interactive Behavior Type
Yelp	19,800	22,734	1,400,002	{Dislike, Neutral, Tip, Like}
Tmall	31,882	31,232	1,451,219	{Page View, Favorite, Cart, Purchase}
Retail	147,894	99,037	1,584,238	{Favorite, Cart, Purchase}

Knowledge-Aware Recommendation Method.

- *KGAT* [35]. It designs an attentive message propagation framework based on knowledge collaboration graph to study higher-order connectivity between semantic relations.
- *KGCL* [53]. It utilizes CL as auxiliary information to combine KG learning with user-item interaction modeling under the joint self-supervised learning paradigm.

Multi-Behavior Recommendation Method.

- *NMTR* [7]. It is a multi-task recommendation framework that can cascade predictions on multiple types of behavior.
- *MATN* [42]. It employs attention networks and memory units to learn interactive dependencies between multiple types of behavior.
- *MBGCN* [13]. It is a graph convolutional network-based model that propagates behavior-aware embeddings on a unified graph by modeling user multiple behaviors.
- *KHGT* [43]. It introduces a transformer-based approach to multi-behavior modeling with emphasis on temporal information and auxiliary knowledge information, modeling behavior embeddings through graph attention networks.
- *CML* [37]. It introduces CL into multi-behavior recommendation, proposing meta-contrastive coding to enable the model to learn personalized behavioral features.

4.3 Parameter Settings

We implement KMCLR with PyTorch and set the users and items embedding sizes of all modules to 32. Embedding initialization is performed in the knowledge enhancement module and the multi-behavior learning module via the Normal and Xavier methods, respectively. We optimize all models by Adma optimizer with the learning rate of $1e^{-3}$ and $3e^{-4}$ for different modules mentioned above. We apply L2 regularization with decaying weights in the range $\{0.05, 0.01, 0.005, 0.001, 0.0005, 0.0001\}$ to learn embeddings. The number of message propagation layers utilized in the model is selected from $\{1, 2, 3, 4\}$. The combination weight α is set to choose from the range $\{0.1, 0.2, 0.3, 0.4\}$. For the compared baseline experiments, the parameters are set according to the original text.

4.4 Performance Validation (RQ1)

We present the performance evaluation results of all baseline methods on different datasets in Table 2 and summarize the following observations:

Compared with all other baselines, KMCLR achieves the best performance on each dataset, which confirms the effectiveness of our model. Since various datasets have large differences in data size and sparsity, it proves that KMCLR has good generality. We mainly attribute the significant improvement on performance to the following two aspects: (1) Benefiting from the introduction of multi-behavior information and additional KG, the model can fully capture personalized information from multiple aspects as signal to guide the recommendation of target behavior. (2) Due to the

Table 2. Performance Comparison on Different Datasets in Terms of HR@10 and NDCG@10

Model \ Dataset	Yelp		Tmall		Retail	
	HR	NDCG	HR	NDCG	HR	NDCG
BPR	0.7435	0.4497	0.2443	0.1501	0.2608	0.1653
LightGCN	0.7884	0.4993	0.3418	0.2048	0.3065	0.1847
AutoRec	0.7592	0.4705	0.3274	0.1903	0.2779	0.1683
NGCF	0.7903	0.5063	0.3288	0.1964	0.3027	0.1844
KGAT	0.8211	0.5362	0.3853	0.2219	0.3485	0.2191
KGCL	0.8331	0.5516	0.4027	0.2293	0.3502	0.2197
NMTR	0.7880	0.4673	0.3623	0.2097	0.3112	0.1869
MATN	0.8210	0.5331	0.4302	0.2437	0.3490	0.2183
MBGCN	0.7962	0.5033	0.3910	0.2261	0.3137	0.1904
KHGT	0.8767	<u>0.5970</u>	0.4032	0.2531	0.4199	0.2471
CML	<u>0.8774</u>	0.5965	<u>0.5185</u>	<u>0.3055</u>	<u>0.4256</u>	<u>0.2503</u>
KMCLR	0.8897	0.6038	0.5671	0.3540	0.4557	0.2735

Bold indicates the best experimental results. The underline indicates the suboptimal experimental results.

Table 3. Results of Ablation Experiments

Model \ Dataset	Yelp		Tmall		Retail	
	HR	NDCG	HR	NDCG	HR	NDCG
w/o-Mcl	0.8352	0.5749	0.4583	0.2645	0.3612	0.2157
w/o-Kcl	0.8687	0.5916	0.5231	0.3064	0.4247	0.2513
w/o NorT	0.8701	0.5943	0.5339	0.3106	0.4268	0.2502
KMCLR	0.8897	0.6038	0.5671	0.3540	0.4557	0.2735

Bold indicates the best experimental results.

introduction of auxiliary self-supervised tasks, the model can excavate individual preferences in terms of higher-level features, clearly distinguishing the information gradient between individuals.

We can notice that the knowledge-aware recommendation and multi-behavior recommendation are always better than the single-behavior recommendation based on CF, which means that it is beneficial to introduce additional auxiliary information into the personalization modeling, which can be to a certain extent making up for the issue of sparse data in a single behavior. In addition, among various baselines, CML is usually the one with the best performance. This result indicates that the introduction of CL in multi-behavior recommendation can make the model better learn the commonalities and differences between individual behaviors, through different semantics guiding more refined recommendations. Furthermore, KHGT also achieves sub-optimal results in most cases, proving the effectiveness of jointly considering multi-behavior views and KG views.

4.5 Model Ablation Study (RQ2)

To explore the impact of different components on performance improvement and demonstrate its effectiveness, we consider three variants of KMCLR and design the following ablation experiments:

- *w/o-Mcl*. In order to study the influence of multi-behavior information on the model, we propose a variant w/o-Mcl, which disables the multi-behavior learning module, and only utilizes the information of a single target behavior for embedding generation and learning.

As can be seen from Table 3, compared with the complete KMCLR, the performance of the version without the multi-behavior learning module is significantly reduced in both indicators. It is because behaviors generated by the same user can profile individual preferences in multiple ways. Utilizing information contrast to model coarse-grained commonalities between different behaviors helps KMCLR learn better representations while making it distinguish fine-grained differences between individuals. The above-mentioned aspects can also be used as supplementary messages to deal with the issue of data sparsity.

- *w/o-Kcl*. To confirm that the introduction of KG is beneficial to enhance the item-side information and plays a positive role in target recommendation, we design another variant *w/o-Kcl*. This variant makes KMCLR degenerate into a simple multi-behavior CL model by disabling the knowledge enhancement module. The experimental results show that the introduction of KG as auxiliary information will improve the performance of the model, which is in line with common sense. Although the introduction of KG may additionally lead to an increase in noisy information, our designed knowledge enhancement module minimizes this effect by selecting stable nodes. Meanwhile, KMCLR processes the KG to generate two graphs with different focus to strengthen the role of knowledge on the final recommendation from different perspectives.
- *w/o NorT*. During the training phase of the model, we propose a joint training paradigm to enhance the learning effect of KMCLR. To verify its effect, we adopt a normal training strategy in ablation experiments, i.e., directly optimizing the linear weighted sum of two contrastive losses and BPR loss. This normal optimization does not consider the integration method between the auxiliary task and the main task, ignoring the impact of different self-supervised learning on the main function. In contrast, KMCLR's strategy is to first train different views' node embeddings individually and then ensemble optimization. The auxiliary information is integrated into the node embeddings from the bottom layer through multiple rounds of optimization. Experimental results show that our training strategy is more conducive to the recommendation of target behavior.

4.6 Performance on Alleviating Data Sparsity (RQ3)

Since the issue of interactive data sparsity is prevalent in real-world scenarios [62], the ability to alleviate this issue becomes a metric for evaluating a recommender system. Additional experiments are conducted to verify the performance of models on interactive data with varying degrees of sparsity. We follow similar settings in [43] to generate datasets of varying degrees of sparsity for Retail. As shown in Figure 3, the *x*-axis indicates the number of user interactions, and the *y*-axis represents the recommendation accuracy.

We conclude that: (1) The CF-based model (NGCF) is greatly affected by data sparsity, so combining additional messages (e.g., multi-behavior information, knowledge information) can mitigate this effect to a certain extent. (2) Compared with simply introducing auxiliary information (KGAT, MBGCN), the model combined with CL (KGCL, CML) is more effective, demonstrating that CL enables the model to maintain the heterogeneity of behavior and the diversity of knowledge semantics through the self-supervised paradigm. (3) Compared with other methods, KMCLR performs consistently when dealing with data with different degrees of sparseness. The results indicate that the performance gap between our approach and other competitors becomes more significant as data become sparser, which proves that the joint introduction of multi-behavior information and KG information can better alleviate the data sparsity, allowing the model to consider multi-view user preferences by incorporating contextual (e.g., click, tag-as-favorite) and auxiliary information.

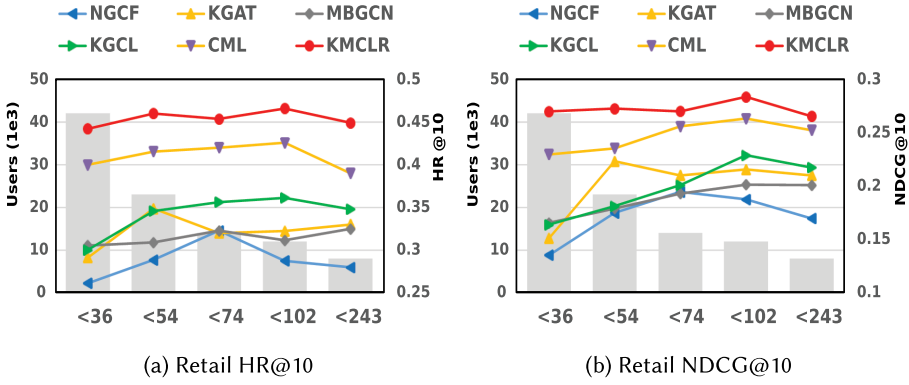


Fig. 3. Performance of KMCLR and baseline methods w.r.t different data sparsity degrees on Retail data.

4.7 Hyperparameter Analyses (RQ4)

To study the effect of hyperparameter settings on KMCLR, we conducted experiments to explore the influence of the hidden state dimensionality d , graph propagation layers L , and the combination weight α on the results under different configurations. The results are shown in Figure 4, where the y-axis represents the degree to which the performance decreases/increases under other settings compared to the best performance with the optimal settings.

- *Hidden State Dimensionality d .* We vary the embedding dimensions in $\{8, 16, 32, 64\}$, and keep other optimal hyper-parameters unchanged. We can observe that the performance of the model improves with the embedding dimension increases from 8 to 32. However, larger embedding dimension does not always cause to stronger model representation, which leads to model overfitting.
- *Graph Propagation Layers L .* By stacking graph propagation layers, the model captures potential dependencies from higher-order neighbors, fully learning the collaboration relationship between higher-order nodes. It can be seen from the experimental results that when L is 3, the model achieves the optimal effect. When stacking more propagation layers further, more noise signals may be introduced, which leads to over-smoothing [23].
- *Combination Weight α .* To distinguish the importance of multi-behavior information and KG information, we set α in the interval $\{0.1, 0.2, 0.3, 0.4\}$. Through the experimental results, it is found that when $0.2 \leq \alpha \leq 0.3$, the model achieves the optimal effect. It proves that too much KG information will add additional noise, while too little will lead to insufficient information enhancement.

4.8 Performance on Alleviating Noise Effect (RQ5)

While the additional auxiliary information is beneficial to improve the recommendation performance, it also leads to the introduction of noise signals. Whether in the multi-behavior interaction graph or KG, there are some invalid nodes that interfere with the final recommendation. Therefore, while using auxiliary information for information enhancement, KMCLR considers adopting methods to reduce the influence of these noise signals. Specifically, for multi-behavior interaction graphs, we set a series of random dropout probabilities to explore optimal sampling strategies under different datasets. By adjusting the interval parameters $[a, b]$ in Equation (15), KMCLR realizes the automatic noise reduction sampling of KG. The interval range reflects the tolerance of the model to the noise signals of KG. We conduct model noise reduction experiments on two

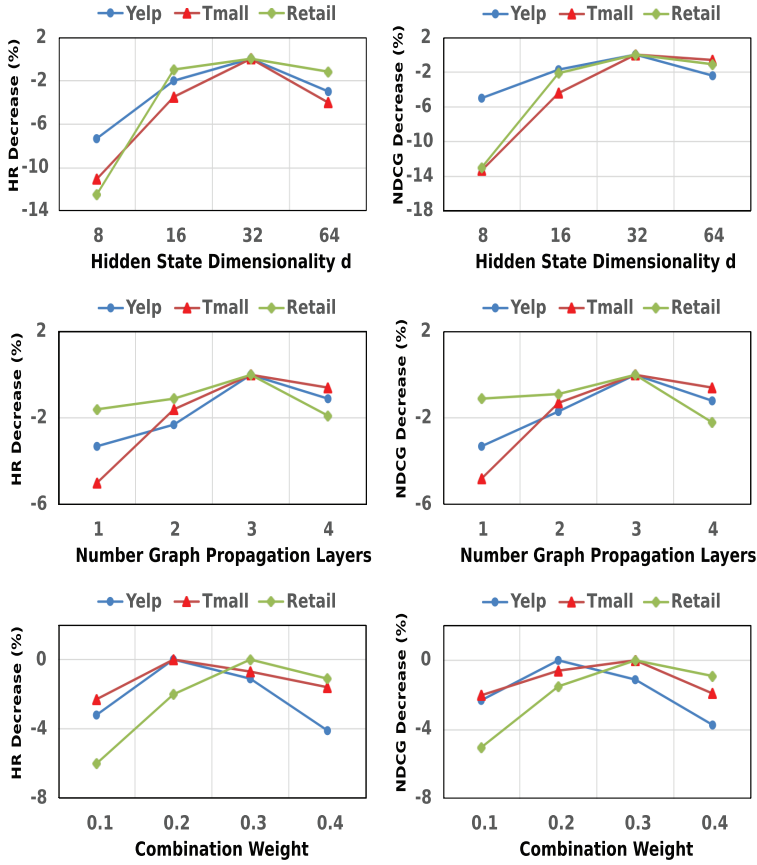


Fig. 4. Hyperparameter analyses of KMCLR.

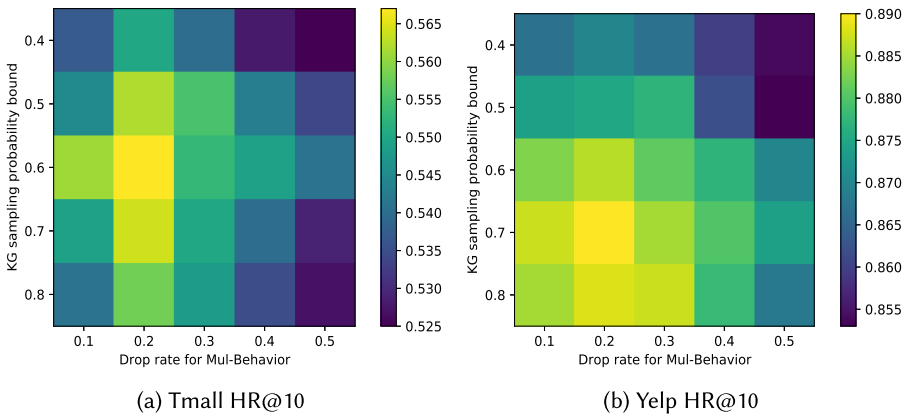


Fig. 5. Parameter settings to alleviate noise.

datasets, and the experimental results are shown in Figure 5. The abscissa represents the random dropout rate of multi-behavior information, and the ordinate indicates the lower limit of knowledge noise tolerance.

Table 4. Results of TMB-Encoder Performance Experiments

Model \ Dataset	Tmall		Retail	
	HR@10	NDCG@10	HR@10	NDCG@10
NARM	0.3421	0.2013	0.3009	0.1816
MA-GNN	0.3547	0.2062	0.3088	0.1887
SR-GNN	0.3603	0.2114	0.3109	0.1901
MATN	0.4302	0.2437	0.3490	0.2183
KHGT	0.4032	0.2531	0.4199	0.2471
CML	0.5185	0.3055	0.4256	0.2503
KMCLR	0.5671	0.3540	0.4557	0.2735
TMBCL	0.5412	0.3386	0.4382	0.2621
KMCLR*	0.5787	0.3602	0.4632	0.2804

Bold indicates the best experimental results.

4.9 Performance on Utilizing Temporal Signals (RQ6)

In order to verify the effectiveness of temporal-aware multi-behavior encoder (abbreviated as TMB-encoder) proposed in Section 3.4, we designed additional verification experiments to show its performance results. Please note that to fully reflect the importance of temporal information and avoid interference from other factors, we replaced $MultiEnc^k(\sim)$ in the KMCLR with TMB-encoder and disabled the knowledge enhancement module. In other words, we modified the KMCLR into a **Temporal-Aware Multi-Behavior Contrastive Recommendation Model (TMBCL)** and conducted comparative experiments with baseline methods from two research lines: temporal-aware recommendation and multi-behavior recommendation. In addition, to comprehensively evaluate the effect of temporal signals on KMCLR, we replaced the multi-behavior encoder in KMCLR with TMB-encoder and abbreviated the new model as KMCLR*. The specific experimental results are shown in Table 4.

Temporal-Aware Recommendation Method.

- *NARM* [16]. It is a recommendation framework based on recurrent neural networks, which utilizes time sequential information and attention mechanisms to construct individual interaction paradigms.
- *MA-GNN* [24]. It distinguishes individual temporal behavior information as short-term and long-term preferences, leveraging memory-augmented graph network to generate corresponding embeddings to extract individual representations.
- *SR-GNN* [40]. It splits the global sequence into session subsequences based on gated technology to explore the feature conversion mechanism between subsequence's nodes.

It can be observed from Table 4 that the model TMBCL and KMCLR* outperform the remaining baseline methods in various indicators on different datasets (except for KMCLR), which proves the effectiveness of the temporal-aware encoder. We consider that due to the introduction of temporal signals, compared to other multi-behavior methods and w/o-Kcl variant (a type of KMCLR variant in 3), TMB-encoder can dynamically captures the evolution of user's preferences, with adjusting the time interval that the model focuses on according to historical interactions, thus emphasizing the modeling of local preferences. Meanwhile, compared with the traditional temporal-aware recommendation method, jointly injecting temporal signals and multi-behavior information into the system can indeed greatly improve the accuracy of recommendations, which allows the model to

depict individual portraits in detail from multiple dimensions and multiple time periods. In addition, the main reason for the performance difference between TMBCL and KMCLR* comes from the introduction of additional KG information, which is reasonable. The KG provides signal enhancement from the item-side to the recommendation system from a different perspective than temporal signals.

4.10 Limitation Analysis

While KMCLR demonstrates superior performance in multi-behavior recommendation scenarios as evidenced by our comprehensive experiments, it is important to acknowledge its inherent limitations regarding multi-modal recommendation contexts. Our framework is fundamentally designed to model behavioral dependencies through CL and KG enhancement, which differs substantially from multi-modal recommendation in both objective and data requirements.

Multi-behavior recommendation focuses on modeling typed interactions (click→favorite→purchase) to capture behavioral dependencies, whereas multi-modal approaches require processing heterogeneous data formats (images, text, videos) with specialized encoders and cross-modal alignment mechanisms. The current architecture of KMCLR lacks the necessary components to effectively encode and align such diverse modal representations, as our embedding spaces are optimized for behavior-specific relational patterns rather than content-based feature extraction. Nevertheless, this limitation does not undermine the significant contributions of our work within the multi-behavior recommendation domain, where KMCLR consistently outperforms state-of-the-art baselines across all evaluation metrics (improving HR@10 by 7.14–9.38% and NDCG@10 by 9.27–15.88% compared to the best baseline), demonstrating its effectiveness in addressing the core challenges of behavioral dependency modeling and target behavior sparsity.

5 Related Work

5.1 Multi-Behavior Recommendation

To enable recommender systems express users' preferences more comprehensively, many methods propose to utilize side information of users/items to represent various relationships [20]. Therefore, recent work has begun to focus on multi-behavior recommendation, attempting to assist the recommendation of target behavior through individual additional behavioral information. For example, work [42] characterizes the relationship between behaviors by designing an attention mechanism. MBGCN [13] uses graph convolutional network to learn the similarity of behaviors to discriminate individual habits. By combining meta-learning with multi-behavior messages, MB-GMN [45] explores hidden embeddings behind behaviors from a low-rank perspective. While our model excavates the inherent semantic relationship by contrasting different behaviors, fully considers the correlation between behavioral cognition and compensates for sparse data in the form of auxiliary signals.

5.2 KG-Based Recommendation

In recent years, KG has become an important data source to enhance the existing recommendation system framework, thus a series of recommendation system methods based on KG have been gradually derived [21, 33, 61]. For example, the classic KGAT model [35] integrates KG information and interaction graph information, performing recommendation prediction through graph attention network. While KHGT [43] attempts to introduce the KG information on the item side into the multi-behavior recommendation and merges it into the entire interactive item representation as a single column of node embeddings. In addition, other methods attempt to utilize KG information to enhance semantic representation, transforming graph information through mapping or other ways, which have also achieved some results [47, 57].

5.3 CL for Recommendation

CL aims to discriminate the similarities and differences between objects through pretext tasks, learning individual characteristics by contrasting positive and negative samples from different views [18, 30, 46, 54]. CL has been demonstrated to be effective in various fields such as computer vision [3], natural language processing [6], and so on. Recently, several works have tried to introduce self-supervised learning into recommender systems and achieved some results [22, 26, 39]. SMIN [22] proposes a heterogeneous graph neural network based on Meta-Relation, which is jointly trained by maximizing the mutual information between local features and global features through self-supervised learning. The work [60] utilizes self-supervised learning for sequence recommendation based on self-attention mechanism. It exploits the correlation of original data to construct self-supervised signals and enhances data representation through pre-training methods to improve sequence recommendation. And our work focuses on exploring behavior semantics and mining knowledge representations, proposing a new CL framework that combines multi-behavior modeling and knowledge modeling.

6 Conclusion

In this work, we propose KMCLR, a novel recommendation framework for multi-behavior CL based on knowledge enhancement. Specifically, we jointly introduce messages of multi-behavior and KG into the recommender system to enhance the information from the user behavior side and item side, alleviating the issue of target-behavior data sparsity. Besides, through designing two CL tasks and a joint training paradigm, KMCLR can learn individual preferences from multiple perspectives, taking into account the commonalities and differences between different behaviors and views. Extensive experimental results on various datasets demonstrate the effectiveness of KMCLR compared to various state-of-the-art methods.

Although KMCLR has achieved robust results, it does not consider temporal signals in feature construction. KMCLR will homogenize data from different time periods, making it difficult to distinguish the user's true preferences in the current period. Therefore, we designed a new multi-behavior recommendation encoder based on temporal information, which can be utilized to capture users' dynamic preferences and behavioral paradigms in different periods to achieve more robust recommendation effects.

Looking ahead, while KMCLR excels in multi-behavior recommendation, it has limitations for multimodal scenarios due to fundamental differences in objectives and data structures. Multi-behavior recommendation models typed interactions, whereas multimodal approaches require specialized encoders for heterogeneous data formats that KMCLR's architecture does not support. Future work will explore bridging these domains through adapter modules and alignment techniques while preserving KMCLR's core strengths in capturing behavioral dependencies and maintaining its superior performance over state-of-the-art methods as demonstrated in our experiments.

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